

SvasthaAI: Futuristic Health OS

An AI-Driven Integrated Healthcare Ecosystem

Technical Documentation & Project Report
2026 Edition

1. Project Overview

SvasthaAI is a comprehensive, AI-driven 'Health OS' designed to bridge the gap between clinical diagnostics, personalized wellness, and accessible healthcare. It serves as a futuristic health ecosystem that integrates deep learning, computer vision, and large language models (LLMs) to provide a holistic health management experience.

Key Objectives:

- Provide real-time, high-accuracy clinical diagnostics for multiple organs.
- Deliver personalized, science-backed wellness coaching (Meal & Workout plans).
- Monitor and analyze mental wellness using multi-modal data.
- Ensure inclusive healthcare access for persons with disabilities.
- Gamify the health journey to ensure long-term consistency.

2. Core Modules

2.1 Multi-Organ AI Diagnostics

A sophisticated computer vision suite utilizing custom CNN models (PyTorch) for analyzing medical scans across five key domains:

- Skin:** Detection of Melanoma, Basal Cell Carcinoma, and Eczema.
- Eye:** Screening for Cataracts, Diabetic Retinopathy, and Glaucoma.
- Oral:** Identifying Dental Caries, Gingivitis, and Ulcers.
- Lungs:** Analyzing X-rays for Pneumonia, Tuberculosis, and COVID-19.
- Bone:** Detecting fractures and signs of Osteoporosis.

2.2 AROMI: AI Wellness Coach

AROMI is a multilingual (English, Hindi, Marathi) health coach that leverages RAG (Retrieval-Augmented Generation) to analyze patient reports and generate 7-day personalized wellness plans.

2.3 ManasMitra: Mental Health Companion

Utilizes 'Tri-Fusion Stress Analysis' (Emotion recognition + App usage + Subjective check-ins) to monitor stress and provide actionable relief strategies.

3. Technical Architecture

Component	Technology Used
Frontend	React 18, TypeScript, Vite, Tailwind CSS, Shadcn UI
Backend	FastAPI (Python), SQLAlchemy, PostgreSQL
AI/ML	PyTorch, OpenCV, Scikit-learn, HuggingFace
LLM & RAG	Groq (Llama-3), LangChain, ChromaDB
Authentication	JWT (JSON Web Tokens), BCrypt
OCR	Python-doctr (Deep Learning OCR)

4. Conclusion

SvasthaAI represents the next generation of healthcare platforms, moving from passive record-keeping to active intelligence. By combining high-speed AI inference with deep personalization, it empowers users to take control of their health destiny.